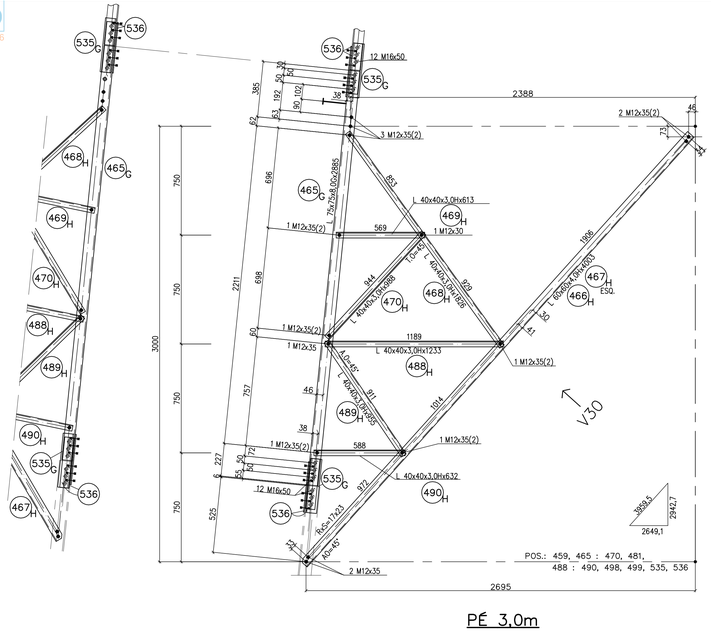
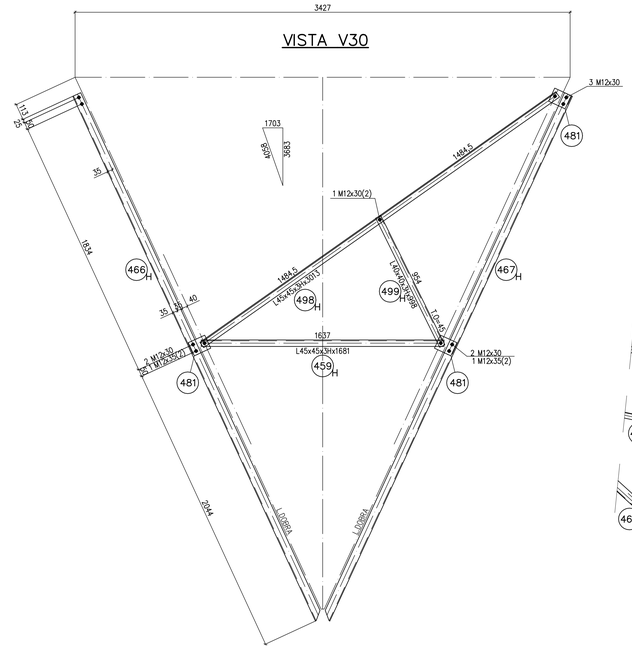
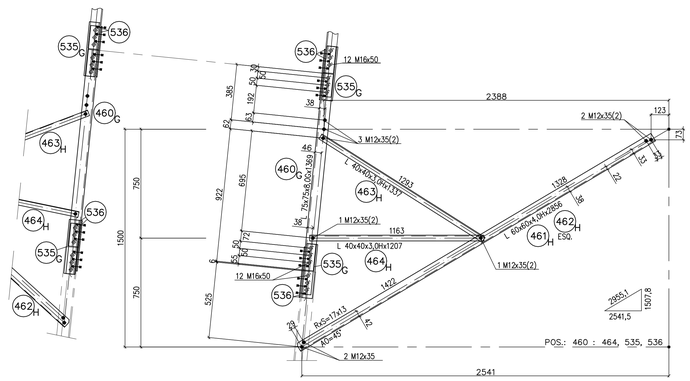
Eletrobras
APROVADO
Por: Wallace Castro em 24/08/2026

PARAFUSOS e ACESSÓRIOS	QUANTIDADES PARA PÉ 1,5m		QUANTIDADES PARA PÉ 3,0m		QUANTIDADES PARA PÉ 4,5m	
PARAFUSOS / COMPRIMENTOS	M12	M16	M12	M16	M12	M16
30	–	–	10	–	27	–
35	18	–	28	–	34	–
40	–	–	–	–	–	–
45	–	–	–	–	–	–
50	–	24	–	24	–	24
55	–	–	–	–	–	–
60	–	–	–	–	–	–
65	–	–	–	–	–	–
70	–	–	–	–	–	–
75	–	–	–	–	–	–
ARRUELA PLANA ESP. 3,0	32	24	61	24	88	24
"PALNUT"	18	24	38	24	61	24
ESPESSURAS						
# 3,0	–	–	–	–	–	–
# 4,0	–	–	–	–	–	–
# 5,0	–	–	–	–	–	–
# 6,0	–	–	–	–	–	–
# 7,0	–	–	–	–	–	–
# 8,0	–	–	–	–	–	–
PARAFUSO DEGRAU PD-18M	–		1		5	

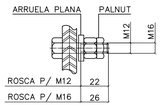


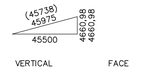
PÉ 3,0m





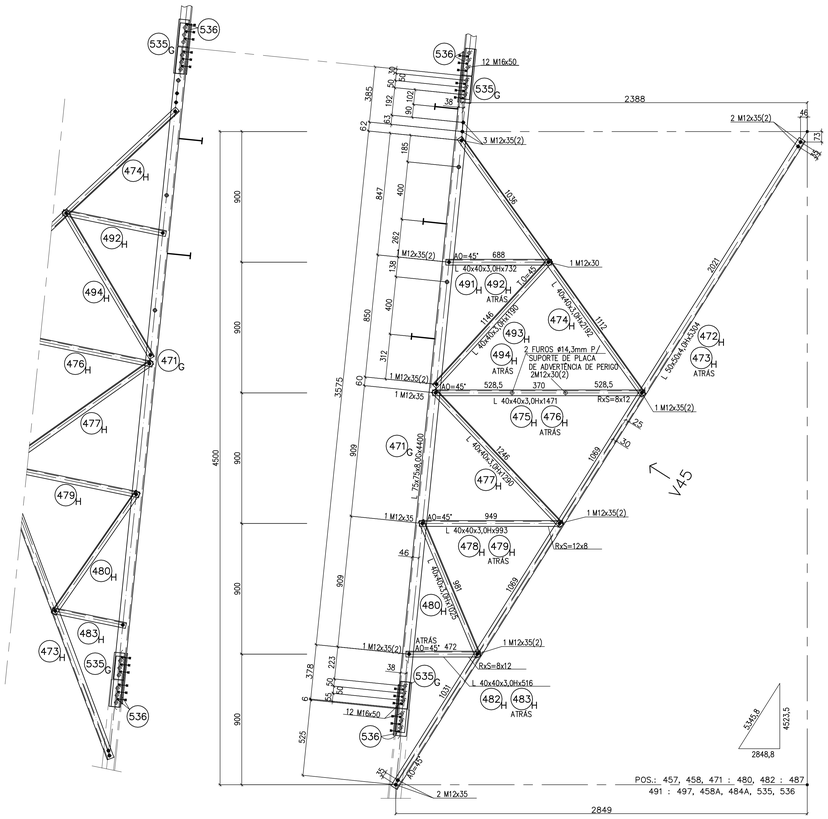
PÉ 1,5m



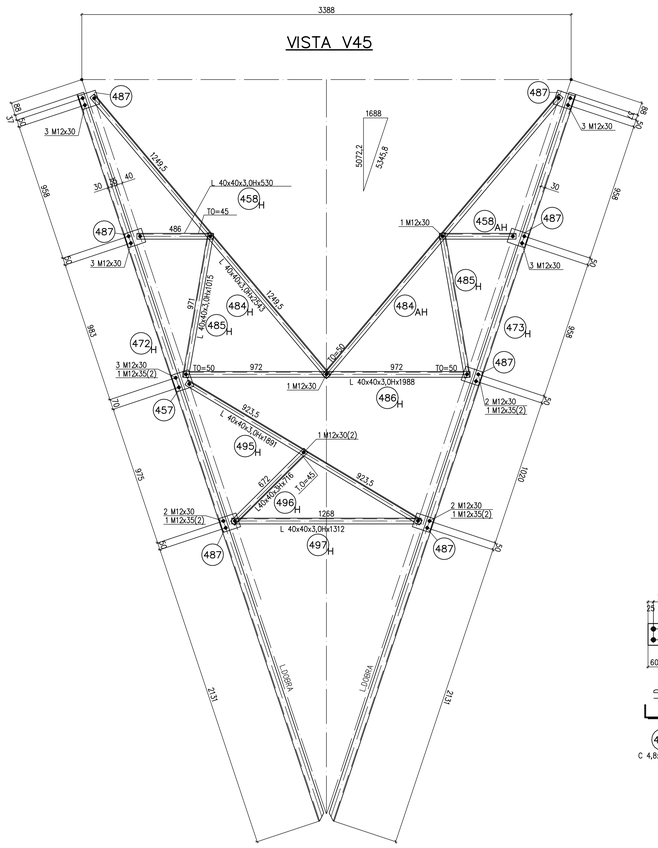


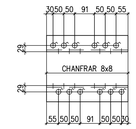
VERTICAL	FACE
tg a = 0,102439	tg g = 0,101906
2tg a = 0,204878	sec g = 1,005179
sec a = 1,005233	

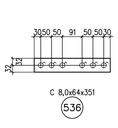
sec b = 1,010439



PÉ 4,5m

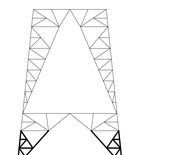


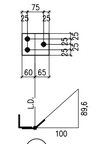


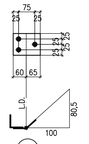












POS.: 535, 536.....BNSM_ADR2451-D2014
POS.: 457 : 499, 458A, 484A

REV.	ALTERAÇÃO	DATA	DES.	VERIF.	APROV.
00	APROVADO	11/08/2026	M.D.R.	L.PENIDO	H.D.G.
00B	CONFORME COMENTÁRIOS	17/07/2026	L.PENIDO	M.D.R.	H.D.G.
00A	EMISSÃO INICIAL	13/03/2026	F.A.F.	L.C.C.	H.D.G.

FINALIDADE

PARA APROVAÇÃO

SIMBOLOGIA DOS PARAFUSOS

- Parafusos M12 – Furos ø13,5 – A=22
- Parafusos M16 – Furos ø17,5 – A=25
- Parafusos M20 – Furos ø21,5 – A=30
- Parafusos M24 – Furos ø26,0 – A=40
- Parafusos Degraus PD16 – Furos ø17,5
- Furos Conforme ø indicado

P/ Treliças

NOMENCLATURA DE RECORTES

RxS (Simples) — (Simples Aba Oposta) — T:0=A (Total Aba Oposta)

RxD (Duplo) — (Duplo Aba Oposta) — (Intermed. Aba Oposta)

TORQUE PARA APERTO DOS PARAFUSOS

ø	Min. (kg.m)	Max. (kg.m)
M12	3,5	4,5
M16	7,5	11,5
M20	13,0	22,0

REFERÊNCIAS

BNSM_ADR2451-D2002 ESPECIFICAÇÕES e NOTAS GERAIS

HENRIQUE GONTIJO
RESPONSÁVEL TÉCNICO
CREA 108.382/MG

LOTE 05 - ANEEL 01/2024

Nº DOCUMENTO: LTRH1.06-230-PE-ME.DMT-0026

Nº EMPREENDIMENTO: NOVA ERA INTEGRAÇÃO

REVISÃO: 00

UNIDADE: LOTE 5 - LT 230 kV

FOLHA: 01/01

LOCAL: BOM NOME - BOM NOME II C1

Nº CONTRATO: -

CÓDIGO INSTALAÇÃO: -

Nº LEGADO: -

TAG: -

TÍTULO: TORRE AUTOPORTANTE DE SUSPENSÃO MÉDIA TIPO BNSM

PÉS: 1,5m, 3,0m e 4,5m

Nº CR GONTIJO: BNSM_ADR2451-D2014

ESCALA: 1/10 - 1/15

REVISÃO: R- 00

Formato A0 (1189x841mm)